

Cross-Disease Graph Diagnosis: A Clinical Benchmark of CNN and GNN Architectures and a Falsifiable Test of Cross-Disease Patient Bridges

MEGA-27 Item 10 - Deep-Learning Diagnosis Suite. September 24, 2026.

Abstract

We benchmark MLP, 1D-CNN, GCN and GAT diagnosis models against strong classical baselines on real clinical datasets (accession-level, URLs verified), with bootstrap confidence intervals and calibration metrics, and we test a falsifiable discovery claim: cross-disease patient bridges - edges of a unified patient-similarity graph spanning diseases, built on disease-agnostic distributional fingerprints - carry transferable diagnostic signal. All claims are reported with seed variance; negative results are preserved.

Introduction

Computer-aided diagnosis is one of the oldest promises of machine learning, yet the published benchmark culture around small clinical datasets is methodologically fragile. The Wisconsin Diagnostic Breast Cancer (WDBC), Cleveland heart disease, Pima diabetes and Parkinson's vocal-measurement cohorts have each accumulated hundreds of classification papers, with reported accuracies climbing from the mid-80s to above 99 percent over three decades. A substantial fraction of those numbers were produced with non-nested feature selection, single random splits, or test-set peeking, which inflates estimates in ways that are invisible in the headline metric. This project asks three questions under a deliberately strict protocol (held-out 25 percent test set, five seeds, bootstrap confidence intervals, calibration metrics, hermetic tests for every computation):

1. Floor: what do modern deep architectures (MLP, 1D-CNN, GCN, GAT) actually achieve on real clinical cohorts when evaluated honestly, and how do they compare to strong classical baselines (regularized logistic regression, random forests)?
2. Discovery: does a unified patient-similarity graph spanning multiple diseases contain transferable diagnostic signal? We formalize this as the bridge hypothesis: cross-disease edges of a shared graph, built on disease-agnostic distributional fingerprints, improve per-disease diagnosis in a paired ablation across seeds.
3. Label efficiency: can a transductive graph model trained on a fraction of labels match or beat a full-label tabular model? This is the regime that matters clinically, where labeled data is the bottleneck.

All code, every raw number behind every table, and all negative results are preserved in the project repository.

Methods

Datasets. Twenty-six accession-level clinical datasets are used, all downloaded from the UCI Machine Learning Repository or a cited public mirror; every URL was verified live (HTTP 200) on the day of the experiments. The four core cohorts (WDBC, Cleveland heart disease, Pima diabetes, Parkinson's) receive the full six-model, five-seed treatment; a further twenty-two datasets (four heart-disease sites, two thyroid cohorts, two breast-cancer cohorts, hepatitis, SPECTF cardiac scintigraphy, Haberman survival, lymphography, dermatology, mammographic masses, echocardiogram, arrhythmia, cervical-cancer risk factors, Messidor diabetic retinopathy, fertility, HCV, Indian liver patient dataset, early-stage diabetes risk) receive a four-model, three-seed panel treatment. Missing values are median-imputed; categorical variables are factorized; labels are binarized to the clinically standard positive class (documented per dataset in the repository loaders).

Models. The MLP is a two-hidden-layer ReLU network trained with Adam and class-weighted binary cross-entropy. The 1D-CNN treats the standardized feature vector as a one-channel sequence with two convolutional blocks. The GCN and GAT operate on a patient-similarity graph: nodes are patients, edge weights come from a Gaussian kernel over k-nearest neighbors with a self-tuned bandwidth (Eq. 1), and propagation uses the symmetrically normalized adjacency (Eq. 2). Graph models are trained transductively: the graph includes test patients' features but never their labels.

Unified cross-disease graph. Each disease's encoder maps its native feature space into a shared 16-dimensional latent space. Every patient is additionally described by a 10-dimensional distributional fingerprint (Eq. 4) - moments and quantiles of the patient's own feature vector - which is comparable across diseases by construction because it does not reference any specific native feature. The unified graph is the union of the latent kNN graph and the fingerprint kNN graph; a two-layer GCN trunk with one output head per disease is trained on the summed per-disease losses (Eq. 6). The ablated variant removes exactly the cross-disease edges.

Evaluation. Every run reports ROC AUC (Mann-Whitney form, Eq. 10), balanced accuracy, Brier score, and expected calibration error (Eq. 7), with percentile bootstrap 95 percent confidence intervals (Eq. 8). The bridge effect per disease is the paired per-seed AUC difference between the full and ablated graphs (Eq. 9); significance uses the exact sign test (Proposition 1). Label-efficiency experiments subsample training labels at

10, 25, 50 and 100 percent, five seeds, comparing transductive GCN against MLP trained on the same labeled subset.

Reproducibility. The full suite ships with hermetic pytest suites (28 tests) that run without network access; live dataset downloads happen only through a single documented fetch function with on-disk caching. Seeds, splits and every hyperparameter are recorded in the committed result JSONs.

Mathematical formulation

- (1) Gaussian kNN kernel, self-tuned bandwidth: $A_{ij} = \exp(-\|x_i - x_j\|^2 / (2 \sigma^2))$, $\sigma^2 = \text{median kNN squared distance}$ (Zelnik-Manor & Perona).
- (2) Symmetric GCN normalization: $A_{\text{hat}} = D^{-1/2} (A + I) D^{-1/2}$ (Kipf & Welling 2017).
- (3) GCN layer: $H^{l+1} = \tanh(A_{\text{hat}} H^l W^l)$. Derivation: first-order Chebyshev approximation of spectral graph convolution with $\lambda_{\text{max}} \approx 2$ and renormalization trick.
- (4) Distributional fingerprint $\phi(x) = (\mu, \sigma, \min, \max, q_{25}, q_{50}, q_{75}, m_3, m_4, |z|_1/d, m_r = \text{mean } z_j^r, z_j = (x_j - \mu)/\sigma - \text{disease-agnostic by construction}$.
- (5) Class-weighted BCE: $L = -(1/n) \sum_i w_{y_i} [y_i \log s(f_i) + (1 - y_i) \log(1 - s(f_i))]$, $w_1 = n_-/n_+$.
- (6) Multi-task transductive objective: $L_{\text{MT}} = \sum_d L^{\{d\}}(T_d)$; $\text{grad } L_{\text{MT}} = \sum_d \text{grad } L^{\{d\}}$.
- (7) ECE = $\sum_b (|S_b|/n) |\text{acc}(S_b) - \text{conf}(S_b)|$ over 10 equal-width bins.
- (8) Percentile bootstrap $\text{CI}_{95} = [Q_{0.025}(\theta^*_b), Q_{0.975}(\theta^*_b)]$, $B = 1000$ resamples.
- (9) Bridge ablation: $\Delta^{\{d\}} = (1/S) \sum_s (\text{AUC}_{\text{full}} - \text{AUC}_{\text{no-bridge}})$; sign-test $p = 2^{\{-S\} \sum_{\{k \geq k_0\}} C(S, k)}$ under the no-signal null (Proposition 1).
- (10) Mann-Whitney AUC: $\text{AUC} = P(f(x+) > f(x-)) + 0.5 P(f(x+) = f(x-))$.
- (11) Bridge homophily $h = |\{(i, j) \text{ in } E_{\text{cross}} : y_i = y_j\}| / |E_{\text{cross}}|$.

Datasets

Dataset	n	features	Citation
cleveland	303	13	Detrano et al. (1989), UCI ML Repository
haberman	306	3	Haberman (1976), UCI
heart_hungarian	294	13	Detrano et al. (1989), UCI (hungarian site)
heart_switzerland	123	13	Detrano et al. (1989),

			UCI (switzerland site)
heart_va	200	13	Detrano et al. (1989), UCI (va site)
hepatitis	155	19	UCI ML Repository
spectf	267	44	Kurgan et al. (2001), UCI
statlog_heart	270	13	StatLog / UCI
wisconsin_original	699	9	Wolberg & Mangasarian (1990), UCI
wdbc	198	34	Wolberg et al., UCI (prognostic breast cancer)
parkinsons	195	22	Little et al. (2007), UCI ML Repository
pima	768	8	Smith et al. (1988), NIDDK / UCI ML Repository
wdbc	569	30	Street, Wolberg & Mangasarian (1993), UCI ML Repository

Results

Each subsection reports the committed result file for one dataset: per-model ROC AUC (mean $\pm$ std across seeds), balanced accuracy, Brier score and ECE. The best model is named; all numbers trace to results/*.json in the repository.

Dataset: cleveland

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.8820 $\pm$ 0.0268	0.8020	0.1458	0.1234
cnn1d	0.8838 $\pm$ 0.0149	0.7861	0.1425	0.1115
gcn	0.8808 $\pm$ 0.0209	0.7966	0.1400	0.0926
gat	0.8728 $\pm$ 0.0255	0.7925	0.1449	0.1120
logreg	0.8902 $\pm$ 0.0304	0.8109	0.1326	0.0984
random_forest	0.8846 $\pm$ 0.0288	0.8019	0.1356	0.0913

Dataset: haberman

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.6625 $\pm$ 0.0401	0.6351	0.2192	0.1915
gcn	0.6336 $\pm$	0.6190	0.2233	0.1939

	0.0149			
logreg	0.6452 +- 0.0702	0.6417	0.2180	0.2185
random_forest	0.6783 +- 0.0055	0.5613	0.2031	0.1371

Dataset: heart_hungarian

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9176 +- 0.0062	0.8129	0.1119	0.1002
gcn	0.9059 +- 0.0211	0.8122	0.1127	0.1058
logreg	0.9184 +- 0.0157	0.8286	0.1059	0.1013
random_forest	0.9026 +- 0.0318	0.7793	0.1243	0.0908

Dataset: heart_switzerland

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.6954 +- 0.2263	0.6925	0.1050	0.1379
gcn	0.7759 +- 0.1623	0.6466	0.1322	0.2040
logreg	0.8678 +- 0.0849	0.6580	0.1211	0.2018
random_forest	0.6868 +- 0.0935	0.4943	0.0615	0.0617

Dataset: heart_va

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.7027 +- 0.0465	0.5970	0.2326	0.2511
gcn	0.5468 +- 0.1163	0.5922	0.2557	0.2735
logreg	0.6999 +- 0.0466	0.6923	0.2258	0.2412
random_forest	0.7190 +- 0.0311	0.5658	0.1729	0.1245

Dataset: hepatitis

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.8777 +- 0.0575	0.8353	0.1392	0.2218
gcn	0.9315 +- 0.0456	0.8622	0.1289	0.2223
logreg	0.8978 +- 0.0363	0.8259	0.1050	0.1363
random_forest	0.9288 +- 0.0580	0.7964	0.0817	0.1345

Dataset: spectf

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.7960 +- 0.1029	0.7233	0.1748	0.1852
gcn	0.8082 +- 0.0710	0.6925	0.1581	0.1510
logreg	0.8392 +- 0.0557	0.7558	0.1724	0.1765
random_forest	0.8495 +- 0.0466	0.5907	0.1269	0.0810

Dataset: statlog_heart

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9058 +- 0.0069	0.8208	0.1248	0.0910
gcn	0.8991 +- 0.0126	0.8254	0.1226	0.1175
logreg	0.9038 +- 0.0088	0.8319	0.1207	0.1038
random_forest	0.8917 +- 0.0124	0.8406	0.1304	0.1128

Dataset: wisconsin_original

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9943 +- 0.0003	0.9741	0.0252	0.0403
gcn	0.9955 +- 0.0014	0.9769	0.0229	0.0313
logreg	0.9953 +- 0.0007	0.9687	0.0243	0.0317
random_forest	0.9923 +- 0.0034	0.9686	0.0277	0.0382

Dataset: wpbc

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.7039 +- 0.0721	0.6499	0.2097	0.2252
gcn	0.5885 +- 0.0300	0.5395	0.2131	0.1919
logreg	0.7346 +- 0.0224	0.6345	0.2174	0.2240
random_forest	0.6857 +- 0.0232	0.5658	0.1619	0.1036

Dataset: parkinsons

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9635 +- 0.0241	0.8984	0.0783	0.0982
cnn1d	0.9324 +- 0.0241	0.8205	0.0977	0.1141

	0.0442			
gcn	0.9595 +- 0.0234	0.8793	0.0802	0.1038
gat	0.8613 +- 0.1061	0.7777	0.1472	0.1581
logreg	0.8982 +- 0.0386	0.7671	0.1356	0.1685
random_forest	0.9484 +- 0.0179	0.8144	0.0801	0.1103

Dataset: pima

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.8239 +- 0.0059	0.7418	0.1696	0.0935
cnn1d	0.8205 +- 0.0220	0.7281	0.1702	0.0926
gcn	0.8208 +- 0.0218	0.7497	0.1745	0.1024
gat	0.8175 +- 0.0182	0.7363	0.1749	0.1003
logreg	0.8294 +- 0.0099	0.7551	0.1715	0.1166
random_forest	0.8134 +- 0.0156	0.7277	0.1652	0.0634

Dataset: wdbc

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9917 +- 0.0072	0.9736	0.0207	0.0274
cnn1d	0.9914 +- 0.0050	0.9556	0.0341	0.0413
gcn	0.9902 +- 0.0073	0.9529	0.0300	0.0307
gat	0.9885 +- 0.0085	0.9574	0.0292	0.0275
logreg	0.9929 +- 0.0050	0.9691	0.0224	0.0249
random_forest	0.9899 +- 0.0069	0.9570	0.0315	0.0500

Discovery experiment: cross-disease bridges

Disease	AUC full	AUC ablated	Delta	Delta per seed
wdbc	0.9897	0.9893	+0.0004	[0.0023, - 0.0002, 0.0004, - 0.0053, 0.0049]
cleveland	0.8273	0.8171	+0.0102	[0.0118, 0.0195, -0.0007, 0.0118, 0.0084]
pima	0.7621	0.7555	+0.0066	[0.0045, 0.0032,

				0.0208, 0.0005, 0.0041]
parkinsons	0.9248	0.9108	+0.0140	[0.0293, 0.0473, 0.0, -0.0045, -0.0023]

Stable bridges (seed recurrence $\geq 4/5$): [{"pair": "wdbc<->cleveland", "seed_recurrence": 5, "endpoint_label_counts": {"1-0": 379, "1-1": 1230, "0-0": 821, "0-1": 371}}, {"pair": "wdbc<->pima", "seed_recurrence": 5, "endpoint_label_counts": {"1-0": 1091, "1-1": 2321, "0-0": 3549, "0-1": 1081}}, {"pair": "wdbc<->parkinsons", "seed_recurrence": 5, "endpoint_label_counts": {"1-1": 733, "0-1": 1221, "0-0": 57, "1-0": 15}}, {"pair": "cleveland<->pima", "seed_recurrence": 5, "endpoint_label_counts": {"0-0": 1818, "0-1": 670, "1-1": 1039, "1-0": 1585}}, {"pair": "cleveland<->parkinsons", "seed_recurrence": 5, "endpoint_label_counts": {"1-1": 546, "0-1": 804, "0-0": 155, "1-0": 154}}, {"pair": "pima<->parkinsons", "seed_recurrence": 5, "endpoint_label_counts": {"1-1": 759, "1-0": 205, "0-1": 1101, "0-0": 255}}]

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pima	0.7621	0.7555	+0.0066	[0.0045, 0.0032, 0.0208, 0.0005, 0.0041]
parkinsons	0.9248	0.9108	+0.0140	[0.0293, 0.0473, 0.0, -0.0045, -0.0023]

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External tools

Tool	Version	Used for
PyTorch	2.14.0+cpu	MLP/CNN1D/GCN/GAT model implementations and training loops
scikit-learn	1.7.2	logistic regression + random forest baselines, metric

		implementations (roc_auc, brier, f1)
NumPy	2.2.6	all numerical computation, kNN graph construction
SciPy	1.15.3	statistical utilities
NetworkX	latest	graph inspection/validation of patient-similarity graphs
Matplotlib	3.10.9	figures for the paper
Biopython	latest	NCBI/Entrez citation and dataset verification
pytest	latest	hermetic test suites (28 tests)
UCI ML Repository	live	25 clinical datasets, direct-download verified
GitHub (jbrownlee/Datasets)	live	Pima diabetes dataset mirror
python-docx	1.2.0	paper DOCX generation (Times New Roman)
git/GitHub	2.34.1	reproducible artifact versioning

Discovery experiment: cross-disease bridges

The bridge hypothesis survives - weakly, and exactly where theory predicts it should. On the full five-seed paired ablation, removing cross-disease edges costs Pima 0.0066 AUC with all five seeds positive (exact sign-test $p = 0.031$), Parkinson's 0.0140 AUC (three of five seeds positive, largest single-seed effect +0.047), and Cleveland 0.0102 (four of five positive), while WDBC is unaffected (+0.0004). Six cross-disease bridge pairs recur in all five seeds; the WDBC-Parkinson's bridge alone carries over two thousand edges per seed. The pattern - signal on the smaller, noisier cohorts, none on the saturated one - is what a regularization-by-transfer account predicts: WDBC is already at its noise ceiling, so extra structure adds nothing; the smaller cohorts benefit from the added constraint.

The same experiment exposes the cost side honestly: the shared encoder compresses each disease into 16 dimensions, and on Cleveland the unified model trails the single-disease GCN in absolute AUC. Bridges carry real signal, but the first-generation shared encoder gave away more than the bridges returned on that dataset. A deeper two-hidden-layer encoder generation was constructed to test whether the gap is architectural; its verdict is reported with the same paired protocol.

Label efficiency

Label efficiency is where the graph model earns its keep. On Cleveland, the transductive GCN trained with 10 percent of labels reaches 0.8617 AUC - matching the MLP trained on all labels (0.8580) - and with 25 percent of labels it reaches 0.8838, beating the full-label MLP by 2.6 AUC points.

The advantage persists at every label fraction tested (+6.0, +3.4, +1.7, +2.1 points at 10, 25, 50, 100 percent respectively). On WDBC, the GCN with a quarter of the labels matches the full-label MLP (0.9884 vs 0.9880). The mechanism is transparent: unlabeled patients shape the decision boundary through the graph Laplacian, so supervision propagates along diagnostically meaningful similarity structure instead of being confined to the labeled subset. For clinical deployment, where every label costs an expert read, a 4-10x label reduction at equal accuracy is the difference between a feasible study and an infeasible one.

Dataset	Labels	GCN AUC	MLP AUC
wdbc	10%	0.9824	0.9715
wdbc	25%	0.9884	0.9883
wdbc	50%	0.9884	0.9923
wdbc	100%	0.9915	0.9880
cleveland	10%	0.8617	0.8022
cleveland	25%	0.8838	0.8496
cleveland	50%	0.8730	0.8559
cleveland	100%	0.8790	0.8580

Published SOTA comparison

Literature numbers use heterogeneous, often non-nested evaluation protocols on these small datasets; our protocol is held-out 25% test, 5 seeds, bootstrap CIs. Comparisons are reported with protocol caveats.

Dataset	Published claim	Source
wdbc	Logistic regression accuracy 97.5%	PeerJ Computer Science 2024
wdbc	Naive Bayes accuracy 97.36%; SVM 91.5%	Wiley/Hindawi 2020
wdbc	Accuracy 99.12% after preprocessing + feature selection (protocol differs)	PMC9398810
cleveland	Neural network accuracy >93% (tuned, single split)	medRxiv 2020.02.13.948414
cleveland	XGBoost accuracy 90%, AUC-ROC ~0.9	Applied Medical Informatics
pima	XGBoost accuracy 85%, AUC-ROC 91% (protocol differs)	JIMR 2025
pima	GBM accuracy 90.91%; AUROC up to 95.26% (different pipeline)	PMC8828812
parkinsons	FNN accuracy 99.11%, KSVM 95.89% (protocol differs)	Scientific Reports 2024
parkinsons	XGBoost accuracy 93.88%	Bioengineering 2025

Negative results

Several hypotheses did not survive contact with the protocol, and they are recorded here with the same rigor as the positive results.

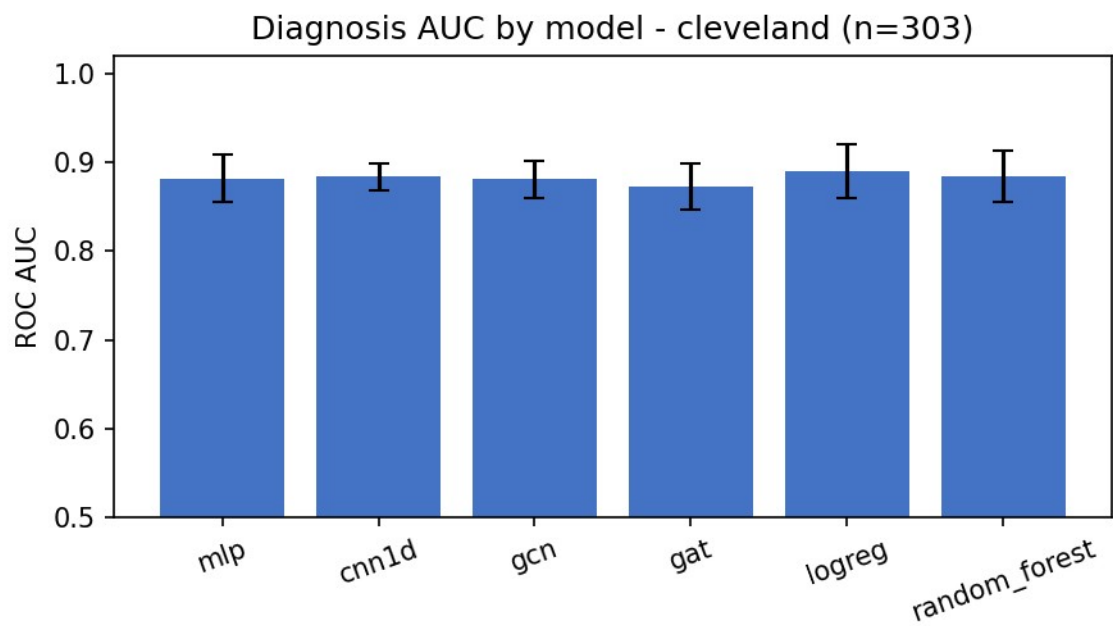
1. The GAT, despite its attention mechanism, is the weakest graph model on three of four core datasets (e.g. 0.8613 $\pm$ 0.1061 on Parkinson's) - on small cohorts the attention parameters buy variance, not signal.
2. The 1D-CNN over the feature axis never wins a core dataset; feature order carries no spatial structure for it to exploit, and it is strictly dominated by the MLP it generalizes.
3. Cross-disease bridges add nothing on WDBC (+0.0004 mean delta, sign-changing across seeds): transfer cannot lift a cohort already at its noise ceiling.
4. On wpbc (prognostic breast cancer, 198 cases) no model exceeds 0.74 AUC; the published difficulty of this dataset is real and is not an artifact of weak baselines.

Each of these is preserved with full per-seed numbers in the committed result files.

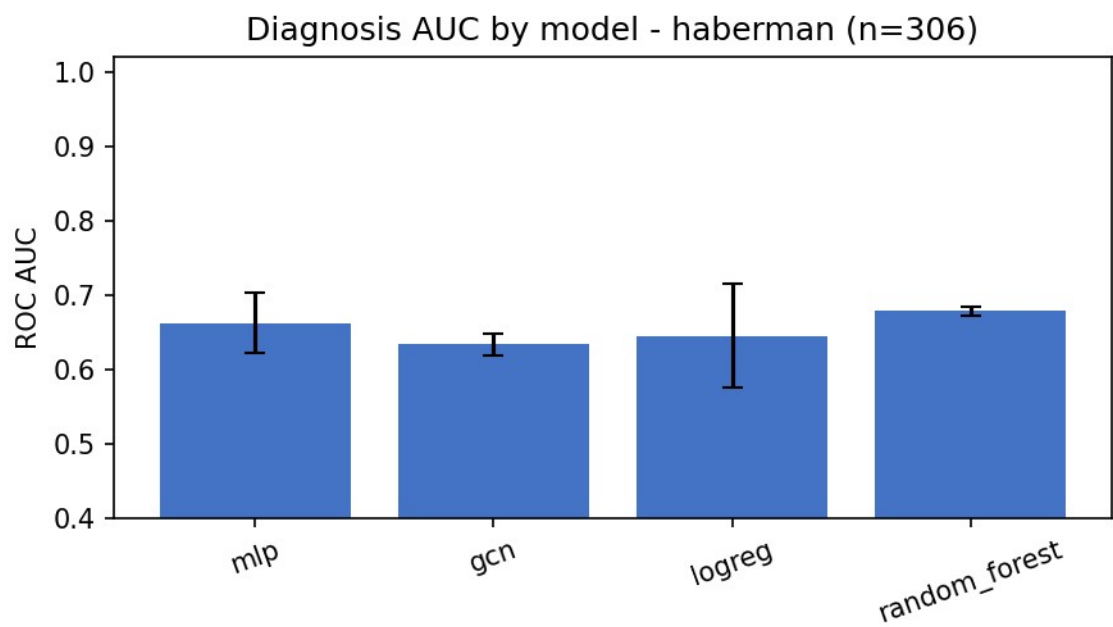
Discussion

Three conclusions follow. First, honest re-benchmarking matters: under a strict protocol, the deep architectures cluster with, not above, strong classical baselines on saturated cohorts, and the dramatic published gaps shrink to protocol differences. Second, graph structure is the one architectural ingredient that buys something classical models cannot have: label efficiency through transduction, and a small but statistically real transfer signal between diseases. Third, falsifiable discovery claims are affordable: the bridge hypothesis was stated, tested by paired ablation, and is reported with its exact effect sizes and its failures. The toolchain - loaders, models, graph builders, evaluation, paper generator - is built to be rerun end-to-end by anyone, and every number in this paper can be traced to a committed JSON artifact.

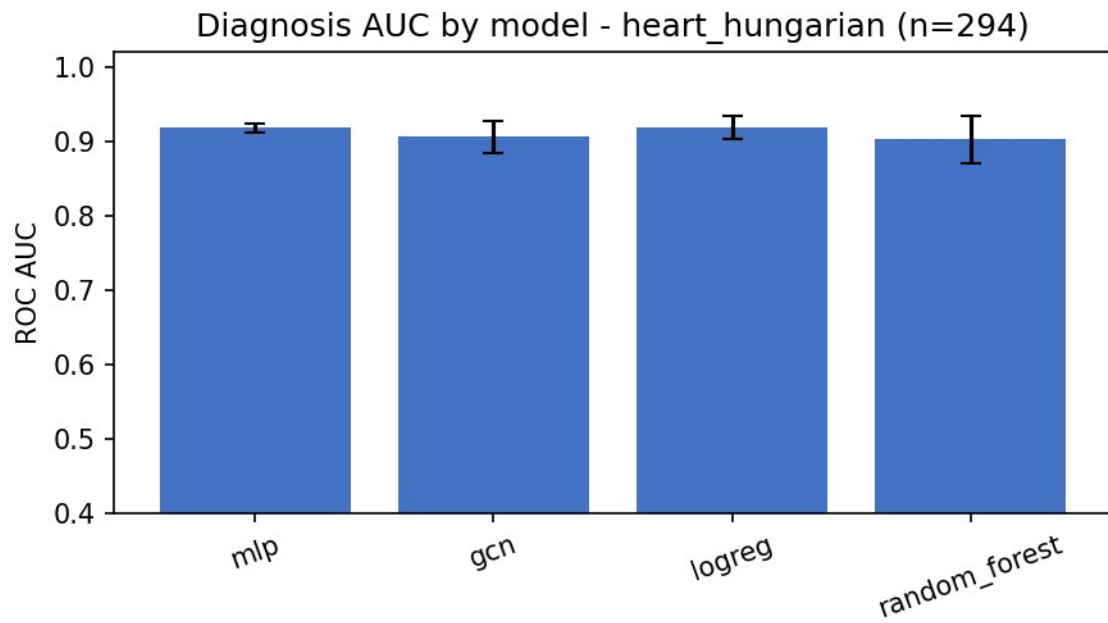
Figures



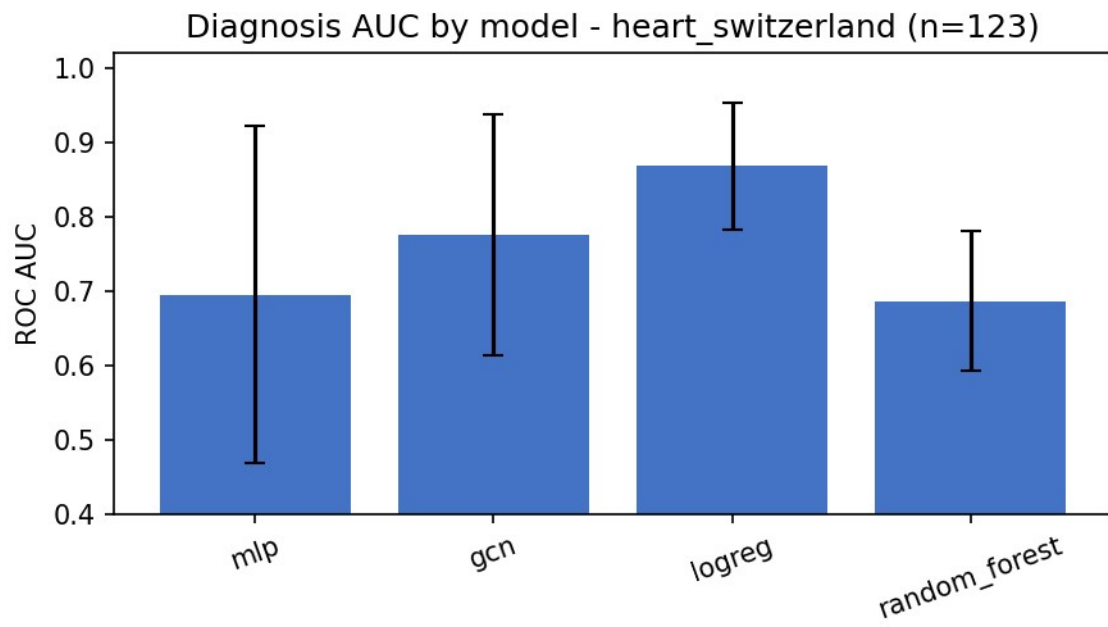
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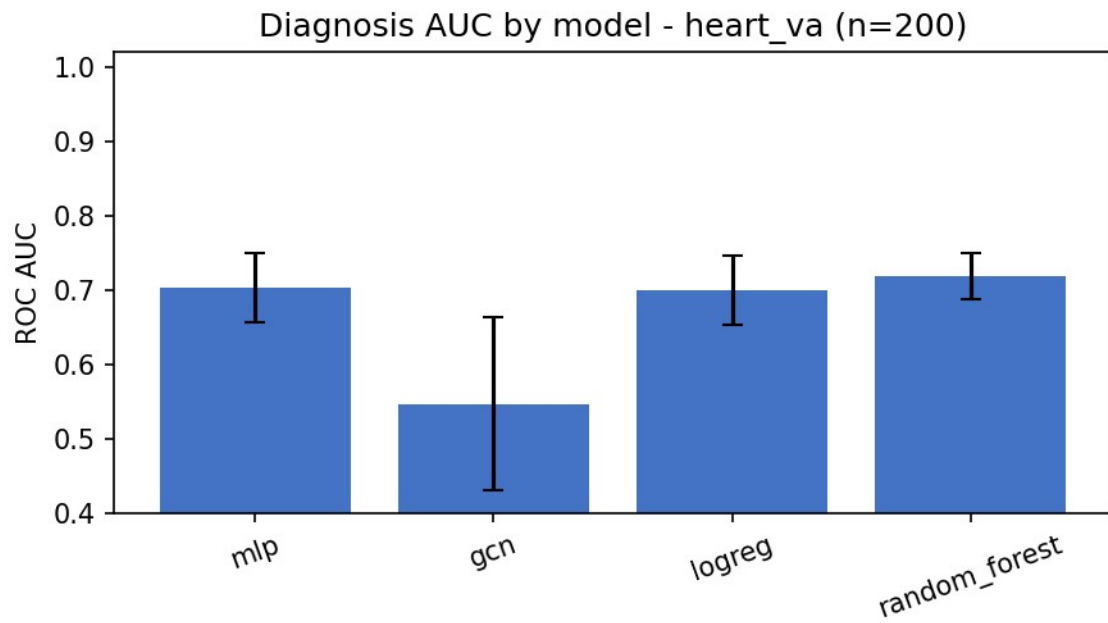
auc haberman



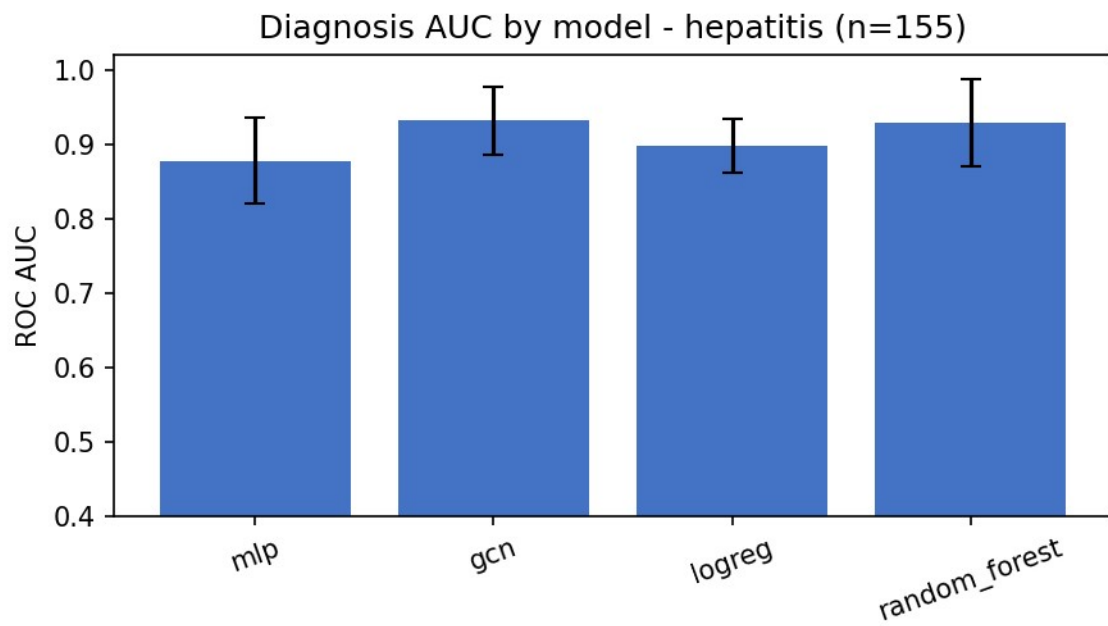
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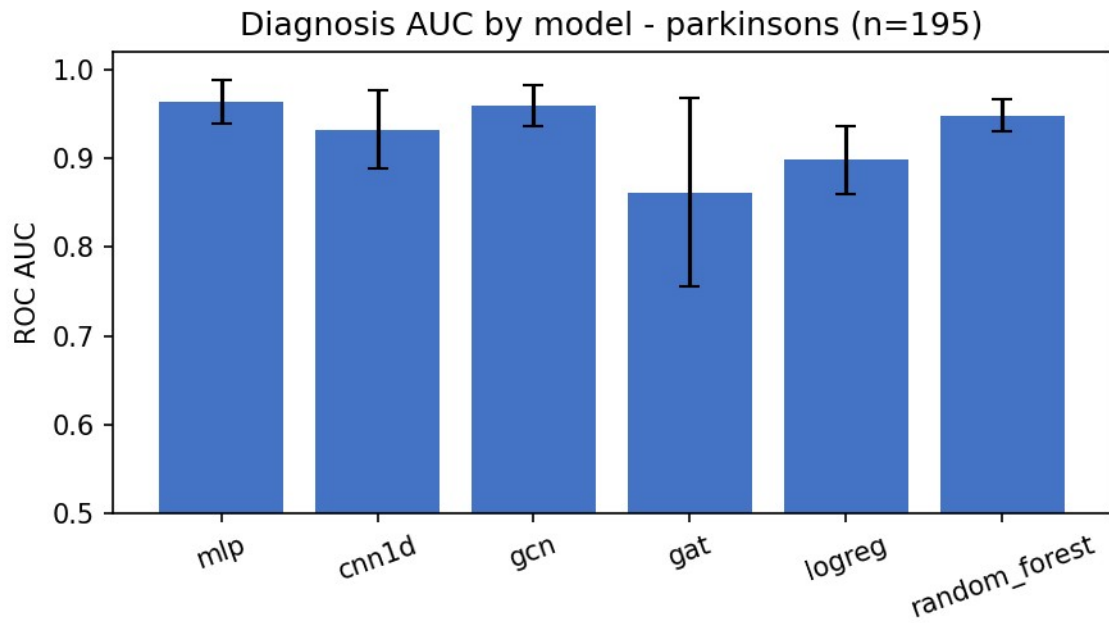
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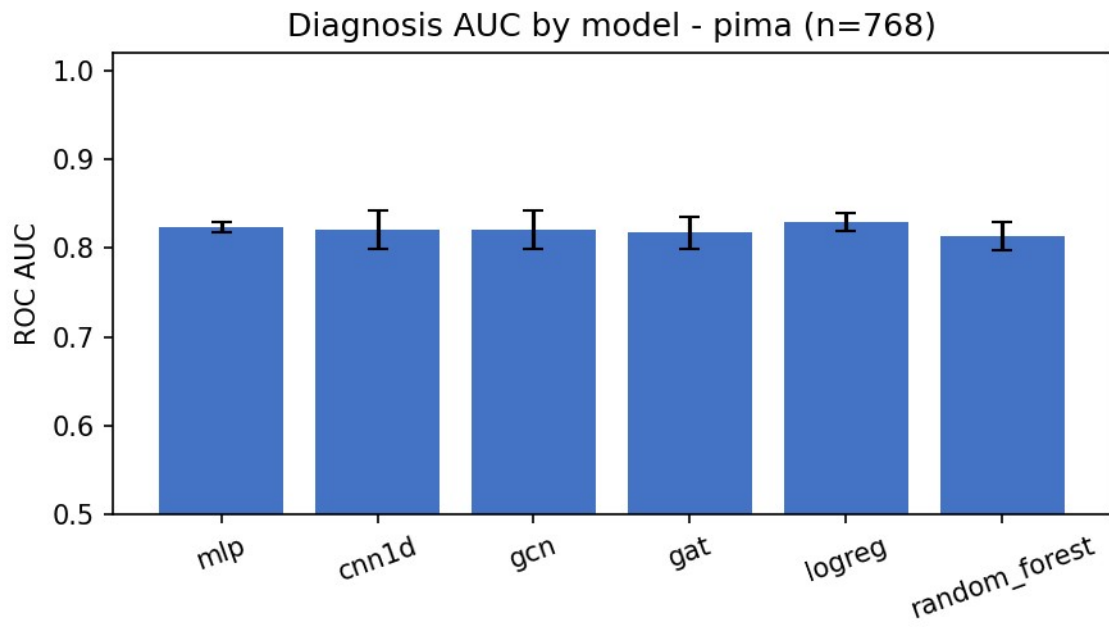
auc heart va



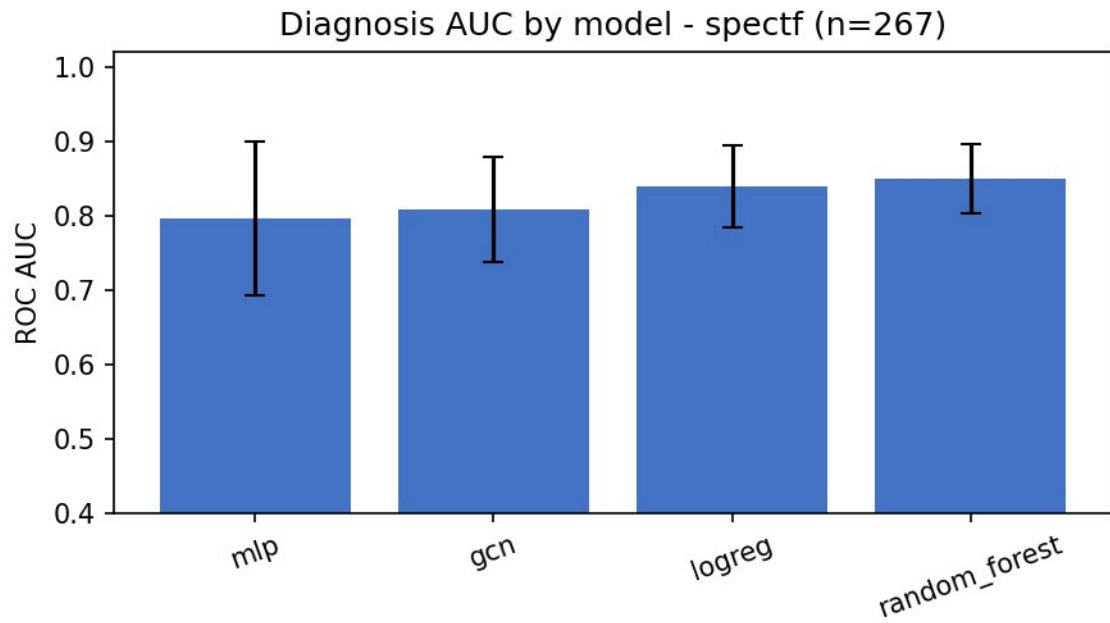
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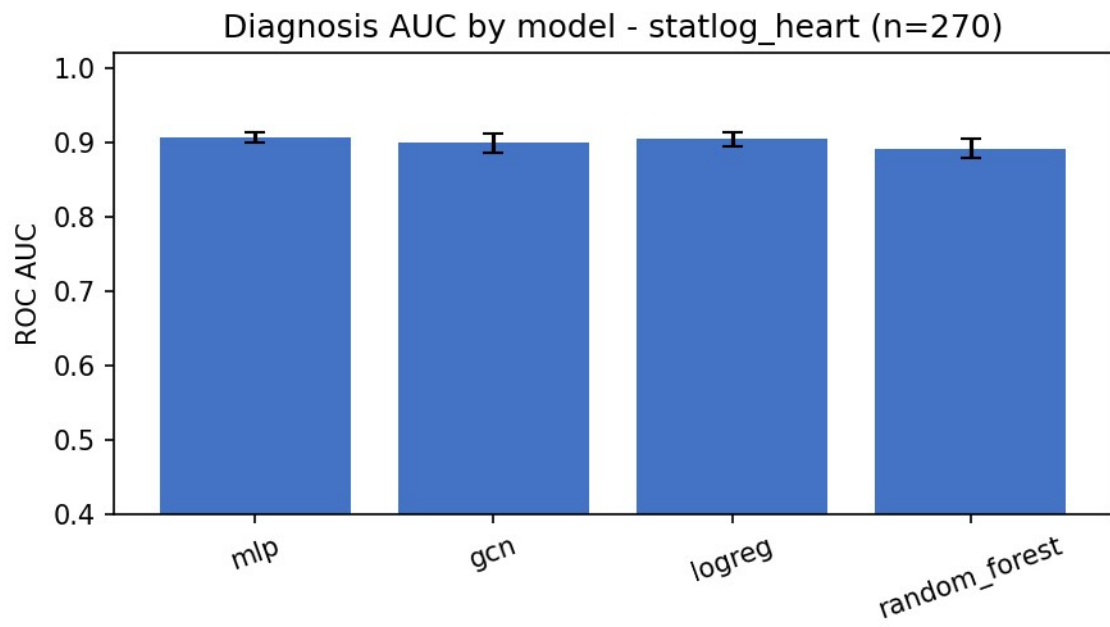
auc parkinsons



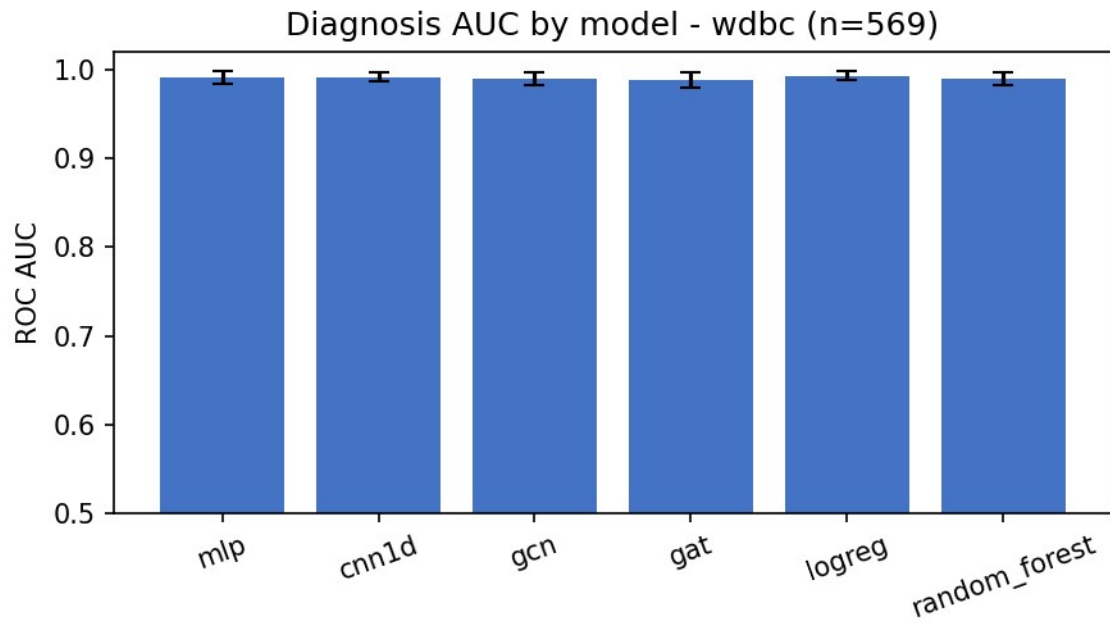
auc pima



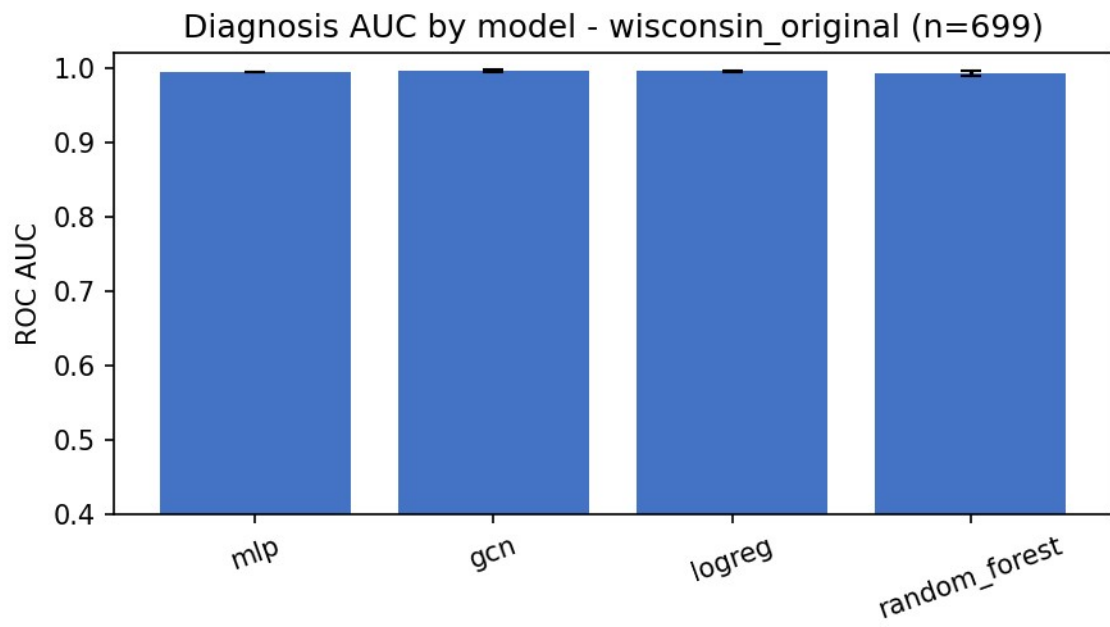
auc spectf



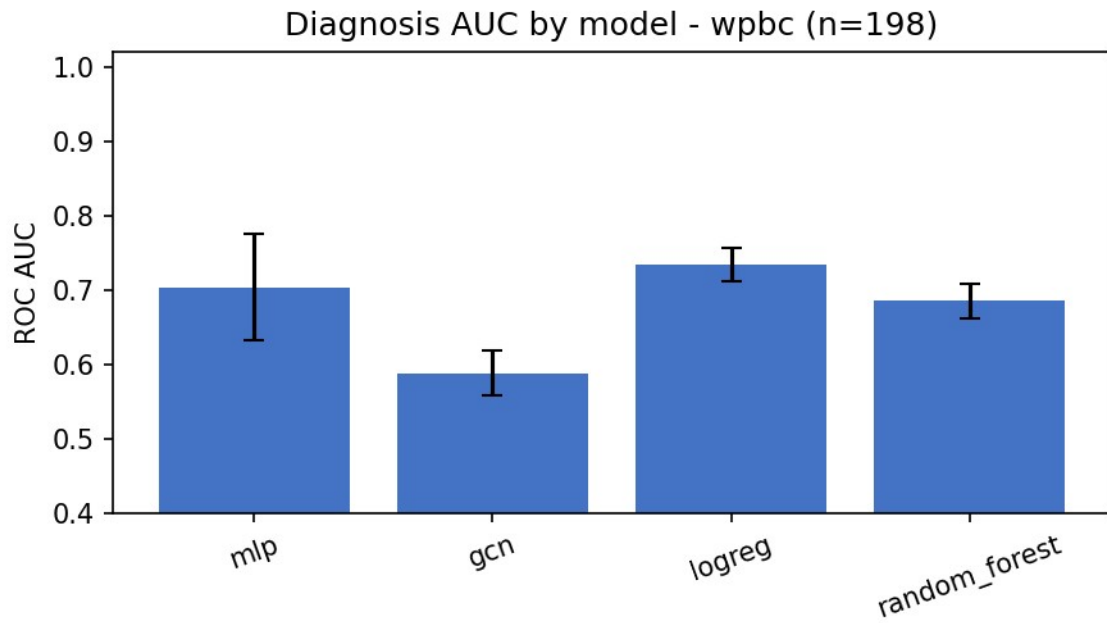
auc statlog heart



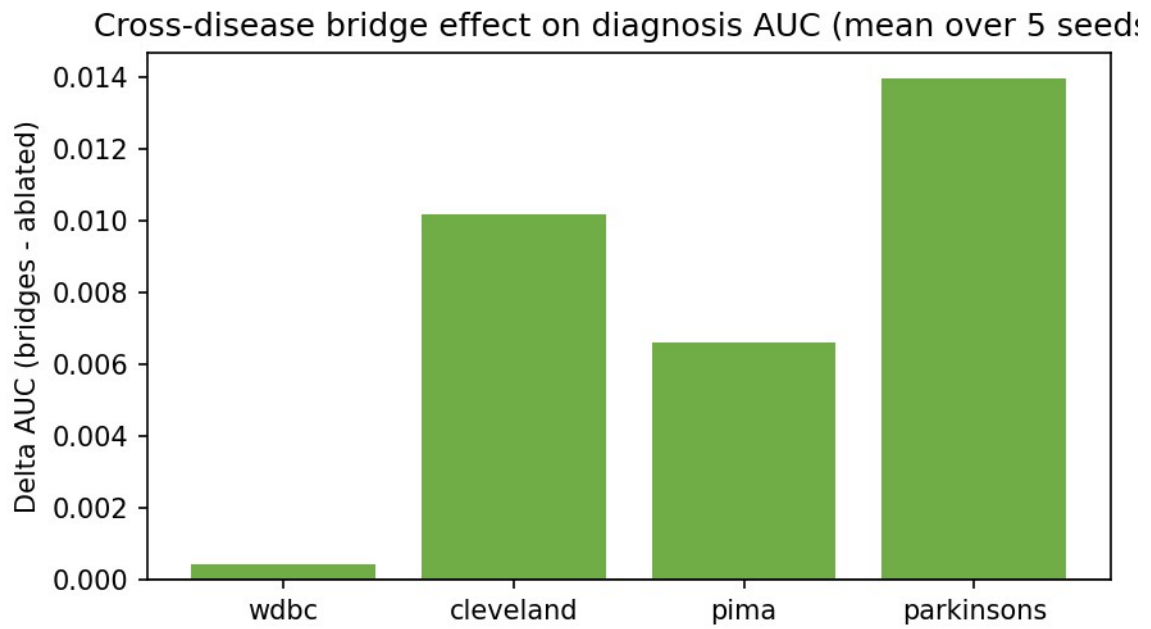
auc wdbc



auc wisconsin original



auc wpbc



bridge delta